Study Report: Fundamentals of Data Engineering - Data Engineer

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Family:	Data Engineer
Level:	Beginner
Chapters:	7
Source:	Reference: Fundamentals of Data Engineering - Joe Reis & Matt Housley

Official sources and free libraries

- https://www.oreilly.com/library/view/fundamentals-of-data/9781098108298/

Summary

Study report for *Fundamentals of Data Engineering* by Joe Reis & Matt Housley (Beginner level) mapped to the Data Engineer role. Modern data stack, pipelines, and team roles in 2024+.

Chapters

Track: Book-Intro

Chapter 1: About Fundamentals of Data Engineering [Beginner]

Chapter focus: About Fundamentals of Data Engineering *(Level: Beginner)**

This study report summarizes *Fundamentals of Data Engineering* by Joe Reis & Matt Housley for the Data Engineer role. The resource is rated Beginner level. Modern data stack, pipelines, and team roles in 2024+. Use this report alongside the official material to prepare for CodeQuest review.

Code Reference:

```
# Fundamentals of Data Engineering
# Author: Joe Reis & Matt Housley
# Role: Data Engineer
# Level: Beginner
```

What it shows: Metadata block records the source book and target role family.

Topic guide

What it actually is
A structured study report based on Fundamentals of Data Engineering.

When to use it
When learning Data Engineer skills at Beginner level.

Where to use it
Data Engineer training paths and certification prep.

Real use example
A learner reads this report before starting the full book or free online guide.

Track: Book-Context

Chapter 2: Why This Book Matters for Your Role [Beginner]

Chapter focus: Why This Book Matters for Your Role** *(Level: Beginner)

Data Engineer professionals use ideas from Fundamentals of Data Engineering to solve real workplace problems. Modern data stack, pipelines, and team roles in 2024+. This chapter explains how the book fits into your learning path and what you should be able to do after studying it.

Code Reference:

```
Role: Data Engineer
Book focus: Modern data stack, pipelines, and team roles in 2024+.
Recommended level: Beginner
```

What it shows: Connects the book topic to the job role outcomes.

Topic guide

What it actually is
Role-aligned learning connects theory to job tasks.

When to use it
During career planning and syllabus design.

Where to use it
Data Engineer bootcamps and CodeQuest teacher assignments.

Real use example
A teacher assigns this book report before a beginner skills checkpoint.

Track: Book-Topics

Chapter 3: Key Topics Covered [Beginner]

Chapter focus: Key Topics Covered *(Level: Beginner)**

The main topics in Fundamentals of Data Engineering include practical concepts described as: Modern data stack, pipelines, and team roles in 2024+. Study each topic with hands-on practice. Take notes on definitions, workflows, and examples that match tools used in Data Engineer jobs today.

Code Reference:

```
- Topic 1: core concept
- Topic 2: core concept
- Topic 3: core concept
- Topic 4: core concept
```

What it shows: Topic list guides what to study chapter-by-chapter in the source.

Topic guide

What it actually is
Topic maps turn a book into actionable learning objectives.

When to use it
Before reading and while building chapter summaries.

Where to use it
Self-study, flipped classroom, and revision.

Real use example
A student maps each book chapter to a CodeQuest report section.

Track: Book-Applied

Chapter 4: Applied: Data Engineer Role Overview [Beginner]

Chapter focus: Data Engineer Role Overview *(Level: Beginner)**

Data engineers build reliable pipelines that move and transform data at scale. You own ingestion, modeling layers, orchestration, and data quality - enabling analysts and ML teams. Cloud-native tooling (managed Spark, serverless warehouses) dominates in 2024-2026.

Code Reference:

```
# Pipeline layers: ingest -> bronze -> silver -> gold -> serve
```

What it shows: Layered architecture separates raw from trusted data.

Topic guide

What it actually is

A data engineer designs and operates data pipelines and platform infrastructure.

When to use it
When data volume, variety, or SLA exceeds spreadsheet/manual scripts.

Where to use it
SaaS analytics, fintech ledgers, ad-tech events, and ML feature stores.

Real use example
CodeQuest nightly exports land in S3; your job makes them queryable by 6 AM for teachers.

Chapter 5: Applied: ETL vs ELT and Batch vs Streaming [Beginner]

Chapter focus: ETL vs ELT and Batch vs Streaming *(Level: Beginner)**

ETL transforms before load; ELT loads raw then transforms in the warehouse (dbt). Batch runs on schedule; streaming processes events continuously. Pick batch unless sub-hour latency is a hard requirement - streaming adds operational complexity.

Code Reference:

```
batch_schedule = '0 2 * * *'  # daily 2 AM UTC
# streaming: micro-batch or continuous with watermarks
```

What it shows: Cron illustrates batch; streaming needs separate ops playbook.

Topic guide

What it actually is
ETL/ELT define where transformation runs; batch/stream define timing.

When to use it
Designing any new data source integration.

Where to use it
Nightly ERP syncs vs clickstream real-time dashboards.

Real use example
Finance needs T+1 batch; fraud team needs 30-second streaming - you run both patterns.

Chapter 6: Applied: SQL for Data Transformations [Beginner]

Chapter focus: SQL for Data Transformations *(Level: Beginner)**

Engineers write heavy SQL: MERGE for upserts, window functions for dedupe, CREATE TABLE AS for snapshots. Use explicit column lists; avoid SELECT *. Version SQL in Git with migration tools.

Code Reference:

```
MERGE INTO curated.orders t
USING staging.orders s ON t.id = s.id
```

```
WHEN MATCHED THEN UPDATE SET amount = s.amount
WHEN NOT MATCHED THEN INSERT (id, amount) VALUES (s.id, s.amount);
```

What it shows: MERGE idempotently applies daily deltas without duplicate rows.

Topic guide

What it actually is
SQL remains the transformation lingua franca in warehouses.

When to use it
Curated layer builds, incremental syncs, and backfills.

Where to use it
Snowflake, BigQuery, Redshift, and Databricks SQL.

Real use example
Staging duplicate keys are resolved with ROW_NUMBER before MERGE - finance totals stay correct.

Track: Book-Plan

Chapter 7: Study Plan and Practice [Beginner]

Chapter focus: Study Plan and Practice *(Level: Beginner)**

Finish Fundamentals of Data Engineering with a weekly plan: read one section, write a summary, complete one exercise, and reflect on how it applies to Data Engineer. If a free edition exists, practice every example. Submit your notes and one mini-project demo for teacher review.

Code Reference:

```
Week 1: Read + notes
Week 2: Exercises
Week 3: Mini project
Week 4: Review + quiz
```

What it shows: A 4-week plan turns reading into demonstrable skill.

Topic guide

What it actually is
Structured study plans improve retention and portfolio outcomes.

When to use it
After finishing the key topics chapters.

Where to use it
CodeQuest end-of-unit assessments.

Real use example

A learner completes the study plan and uploads a beginner project screenshot.